# Mohamed Alhaskir

AI Researcher | Research Engineering, Multimodal AI

AI researcher and PhD candidate at RWTH Aachen University with experience designing, implementing, and evaluating end-to-end AI systems. Research spans multimodal LLM applications, deep learning for physiological time series, and predictive modeling. Experienced in reproducible ML experimentation, model evaluation, GPU-based research workflows, and developing AI systems for real-world applications.

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## RESEARCH & ENGINEERING EXPERIENCE

**PhD Researcher**, Institute of Medical Informatics, RWTH Aachen University 2022 – Present

*Aachen, Germany*

**HaMoJo: LLM and Computer Vision System for Clinical Simulation Assessment** | github.com/Mohamed-Alhaskir/HaMoJo

- Architected and implemented an end-to-end multimodal AI pipeline integrating speech recognition, speaker diarization, computer vision, and locally hosted LLMs to generate structured feedback from recordings of medical students' simulation training.
- Developed reproducible evaluation workflows incorporating rubric-grounded LLM scoring, deterministic inference, and comparison against human assessments.

**MoniPy: Deep Learning for Wearable ECG Seizure Detection** | github.com/rwth-imi/monipy

- Designed and implemented a deep learning pipeline combining physiological signal processing, heart-rate variability features, automated data-quality assessment, and seizure classification.

**FORECAST: Foundation Models for Hospital Patient Flow Prediction** (ongoing)

- Applying foundation models to routine emergency department data to predict inpatient ward admission and subsequent length of stay for resource planning.

**Master's Researcher**, Forschungszentrum Jülich (IBI-3) 2021 – 2022

- Master's thesis on spike-waveform variability in 3D microelectrode array recordings of neuronal cultures; developed signal-processing algorithms and optimized the SpyKING CIRCUS spike-sorting pipeline.

**Research Internships**, Charité Berlin and RWTH Aachen 2020 – 2021

- Charité Berlin, AG Klinische Neurotechnologie: signal-acquisition and decoding algorithms for an OPM-MEG-based brain-computer interface.
- RWTH Aachen, Molecular & Systemic Neurophysiology: classification algorithms for neural decoding of wide-field calcium-imaging data.

## TECHNICAL SKILLS

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|---|---|
| **Programming** | Python, SQL, Bash, MATLAB |
| **Machine Learning** | PyTorch, TensorFlow/Keras, scikit-learn, Hugging Face Transformers, LoRA fine-tuning |
| **LLMs & Multimodal AI** | vLLM, llama.cpp, Whisper, speaker diarization, MediaPipe, OpenCV |
| **ML Evaluation** | Model performance evaluation, reproducible experimentation, data-quality assessment |
| **ML Engineering** | Git, pytest, Docker, MLflow, Weights & Biases, FastAPI |
| **Computing** | Slurm, HPC environments, GPU-based experimentation |
| **Data & Statistics** | pandas, NumPy, SciPy, statsmodels, SQLAlchemy/MySQL |
| **Languages** | Arabic (native), German (C2), English (C1) |

## PUBLICATIONS

### Under Review

**Alhaskir, M.**, et al. *Trustworthy Artificial Intelligence: Seizure Detection by Wearables Following International Consensus Guidelines.* 2026.

**Alhaskir, M.**, et al. *Automated Feedback Generation in Paediatric Simulation Training: A Prospective Validation of AI-Generated Structured Feedback Reports.* 2026.

### Journal Articles and Preprints

**Alhaskir, M.**, et al. *Reliable Detection of Focal Onset Impaired Awareness Seizures in Patients with Epilepsy Using Wearable ECG: Development and Validation Study.* Computer Methods and Programs in Biomedicine, 279, 109295, 2026.

Tayebi Arasteh, S., Farajiamiri, M., Lotfinia, M., Hinrichs-Puladi, B., Bienzeisler, J., **Alhaskir, M.**, et al. *The Role of Self-Supervised Pretraining in Differentially Private Medical Image Analysis.* arXiv:2601.19618, 2026.

Bienzeisler, J., Hertwig, M. K., Heidemeyer, H., **Alhaskir, M.**, et al. *Trans-Sectoral Patient Pathways in Urgent and Emergency Care (TRANSPARENT Study): Protocol for a Prospective, Mixed-Methods Study in Germany.* BMJ Open, 16(2), e114590, 2026.

### Conference Papers

**Alhaskir, M.**, et al. *Feasibility of LLM-Assisted Communication Feedback in Simulation-Based Medical Education: A Pilot Study.* Studies in Health Technology and Informatics, 336, 2140–2144, 2026.

Bienzeisler, J., Heidemeyer, H., Röhrig, R., Kombeiz, A., **Alhaskir, M.**, et al. *A Regional Data Space for Urgent and Emergency Care in Germany.* Studies in Health Technology and Informatics, 336, 1167–1171, 2026.

Bienzeisler, J., Hoy, W., Kombeiz, A., **Alhaskir, M.**, et al. *Beating the Clock: A Prediction Model for Timely Care in Emergency Departments.* Studies in Health Technology and Informatics, 316, 1657–1658, 2024.

**Alhaskir, M.**, et al. *ECG Matching: An Approach to Synchronize ECG Datasets for Data Quality Comparisons.* Studies in Health Technology and Informatics, 307, 225–232, 2023.

**Alhaskir, M.**, et al. *Spectral Fusion of Heartbeat and Accelerometer Data for Estimation of Breathing Rate in Wearable Patches.* Studies in Health Technology and Informatics, 302, 1025–1026, 2023.

## EDUCATION

**PhD, Medical AI**, RWTH Aachen University — 2022 – 2027 (expected)
*Institute of Medical Informatics · Aachen, Germany*

**MSc, Biological Information Processing**, RWTH Aachen University — 2019 – 2022
*Major: Biophysics · Aachen, Germany*

**BSc, Biological Information Processing**, RWTH Aachen University — 2016 – 2019
*Aachen, Germany*

## AWARDS & LEADERSHIP

DAAD Prize for Outstanding International Students, RWTH Aachen, 2021

**Co-Founder, NeuroTX Aachen e.V., 2020–2022**

Developed a brain-computer interface prototype for electric wheelchair control; project funded by the RWTH Aachen Collective Incubator.